# Water-Wave Dysthe: Research Outline

**Abstract**

Can a learned model predict unseen water-wave packets accurately and at useful cost? This outline fixes the scope and evidence requirements. Reference migration and learning experiments remain pending; no new result is claimed.

## 1 Water-wave model

The baseline is the spatial Dysthe equation for a weakly nonlinear, narrowband, unidirectional deep-water free-surface envelope $u(\xi, \tau) \in \mathbb{C}$:

$$\begin{aligned} u_\xi = & -iu_{\tau\tau} - i|u|^2 u \\ & - \epsilon\big(8|u|^2 u_\tau + 2u^2\overline{u_\tau} + 2iu|D_\tau|(|u|^2)\big). \end{aligned} \tag{1.1}$$

Here $\xi$ is propagation distance, $\tau$ is retarded time, $\epsilon$ is carrier steepness, and $|D_\tau|$ has Fourier multiplier $|k|$. This is the free-surface convention of Fedele and Dutykh, equation (1.3). The shared contract is `water-wave-dysthe-spatial-v1`. The computational profile is periodic; there is no resolved transverse direction. Other physical settings are excluded from this research.

## 2 Reference and learning protocol

Verify the existing water-wave reference, including signs, mean flow and the NLS limit. Qualify packet families by grid, propagation-step and domain refinement. Compare an FNO baseline with a hybrid numerical–learned stepper. Record losses, normalization, optimizer, validation-rollout early stopping, checkpoints, seeds and hardware. Keep each initial condition and its refinements in one split; reserve complete test trajectories. Physics penalties must follow (1.1). The raw envelope is noncanonical: a Hamiltonian constraint requires the appropriate transformation and boundary convention.

## 3 Evidence and publication

Measure weighted $L^p$ errors over $\tau$, phase, modulation growth, focusing distance, invariant drift and accumulated propagation error. Compare inference cost at matched accuracy and report reference-generation and training costs separately. Include ablations, failures and variability across cases and seeds. Conservation alone does not establish field or phase accuracy.

Choose one demonstrated contribution and compare it with the closest literature. Link each claim to configurations, source revisions, artifact locations and hashes. Preserve compact figure data and vector PDF figures. The evidence ledger is empty. Historical results from other physical models must not be relabeled or used to support this paper.